

Dual-Branch Multi-Granularity Network with Structured Contrastive Ranking for Cross-Modal Retrieval

Zihao Chen
zhchen@mail.hfut.edu.cn
The Key Laboratory of
Knowledge Engineering
with Big Data (the Ministry
of Education of China),
Hefei University of
Technology
Hefei, Anhui, China

Chenyang Bu
chenyangbu@hfut.edu.cn
The Key Laboratory of
Knowledge Engineering
with Big Data (the Ministry
of Education of China),
Hefei University of
Technology
Hefei, Anhui, China

Shengwei Ji
jisw@hfu.edu.cn
School of Artificial
Intelligence and Big Data,
Hefei University
Hefei, Anhui, China

Xindong Wu*
xwu@hfut.edu.cn
The Key Laboratory of
Knowledge Engineering
with Big Data (the Ministry
of Education of China),
Hefei University of
Technology
Hefei, Anhui, China

Abstract

Cross-modal retrieval (CMR) has advanced considerably by mapping image and text features into a shared embedding space; however, these approaches still face two persistent challenges: (1) **semantic sparsity**, where discriminative cues are confined to localized regions, making it difficult to identify implicit visual evidence; and (2) **ranking uncertainty under semantic ambiguity**, where models struggle to maintain the correct retrieval order when candidates share similar contexts. To address these issues, we propose the *Dual-Branch Multi-Granularity Network (DBMG)* with *Structured Contrastive Ranking*, which enriches visual semantics by leveraging a multimodal large language model to generate auxiliary descriptions, aligns sparse cues through a dual-branch architecture capturing both global and local interactions, and enforces ranking consistency via a three-stage contrastive objective that progressively optimizes category clustering, instance alignment, and margin-based ranking. Extensive experiments on four standard CMR benchmarks demonstrate that DBMG outperforms 12 strong baselines, achieving an average 15.91% improvement in mAP, establishing a new state-of-the-art. The code is available at <https://github.com/DMiC-Lab-HFUT/DBMG>.

CCS Concepts

• Information systems → Retrieval models and ranking.

Keywords

Cross-modal retrieval, Contrastive learning, Multimodal alignment

ACM Reference Format:

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*Corresponding author.



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1 Introduction

With the rapid advancement of multimodal Retrieval-Augmented Generation (RAG) [1, 34, 43] and GraphRAG systems [17, 39, 47, 53, 54], CMR has become increasingly important for gathering visual and textual evidence to support reasoning in large models. However, images often contain rich and complex information that is difficult to annotate exhaustively in advance [2]. Constructing comprehensive multimodal knowledge graphs [3], for example, is not only costly but also impractical when user intents are unknown a priori [3]. This leads to a *critical challenge*: *how can we effectively retrieve relevant multimodal content based on user queries when the discriminative evidence is localized in specific image regions, without relying on extensive pre-annotations?*

Existing CMR methods fall into two main categories. The first learns a shared embedding space to align global representations of images and texts [10, 40]. While efficient, these methods typically miss fine-grained correspondences when discriminative cues are localized. For instance, consider the query “Apple and Nvidia are collaborating” paired with an image from Figure 1 that depicts office scenes from both companies. Although the image contains abundant visual information—such as desks, employees, staircases, and general office environments—the truly discriminative evidence for accurately matching the query is confined to the specific regions containing the Apple and NVIDIA logos. A global embedding method, which encodes the entire image into a single representation, is likely to be overwhelmed by such redundant visual data.

The second line of work leverages local information for fine-grained modeling [6]. Several strategies have been developed under this paradigm: (1) Object detection-based methods [20, 21, 37] employ pre-trained detectors to identify salient objects or regions in images and align them with relevant textual phrases. While these methods improve localization, they are inherently constrained by the predefined categories of the detection models, and often fail to recognize fine-grained or out-of-domain visual concepts that are critical for accurate retrieval. (2) OCR-based techniques [13, 26] extract textual content embedded within images, such as signs, logos, or captions, and incorporate them as additional semantic clues. However, these methods are limited to recognizing textual information and cannot interpret purely visual elements. (3) Local attention mechanisms [38] dynamically weigh the importance of different image regions or text tokens during cross-modal interaction. Though flexible, these approaches often lack explicit grounding and

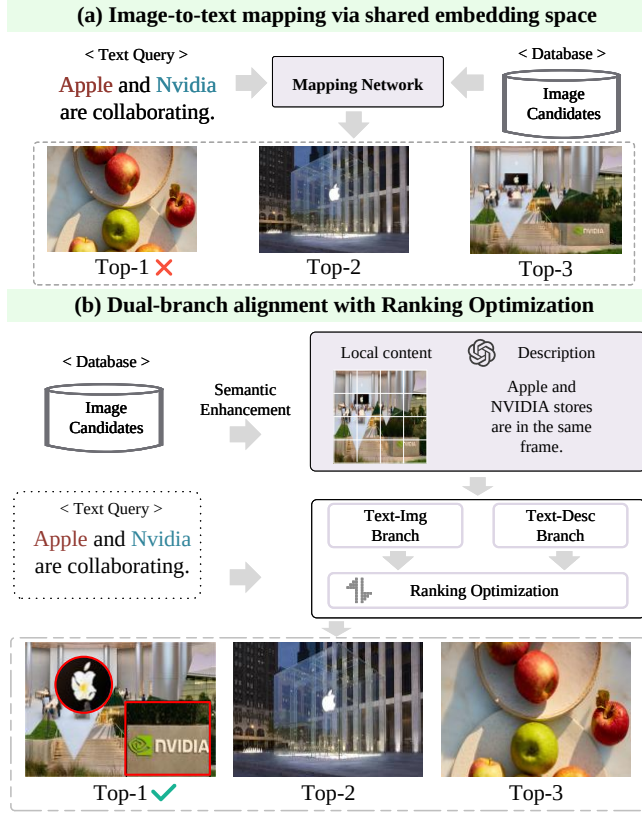


Figure 1: Illustration of our motivation. (a) Conventional global image-text mapping via a shared embedding space. (b) Our dual-branch alignment with semantic enhancement and structured ranking optimization, to capture localized discriminative cues.

may not reliably focus on semantically meaningful parts, especially when fine-grained cues are implicit. (4) Multi-granularity networks integrate features from both global and local representations to capture hierarchical semantic structures. Nevertheless, without incorporating recognition or OCR modules, such methods struggle to interpret localized regions beyond vector-level similarity. For instance, as shown in Figure 1, even if the correct image region is highlighted, the model may fail to recognize specific concepts such as “NVIDIA” without explicit semantic guidance, thereby limiting retrieval accuracy.

The above analysis highlights two fundamental challenges in current CMR methods: (1) **Semantic sparsity**. Discriminative information is often sparsely distributed in local regions of images or specific tokens in text, while the majority of content remains semantically redundant. This sparse distribution makes it difficult for models to identify and correlate the truly relevant cues, especially when critical evidence is implicitly embedded in the visual scene—such as the NVIDIA logo in a corporate office image—rather than explicitly mentioned in the paired text. (2) **Ranking inconsistency from ambiguous semantics**. While multi-granularity modeling helps identify sparse local cues, it alone cannot ensure consistent ranking when candidates share similar visual or textual

context. For instance, images of “apple” (fruit) and “Apple” (brand) may both contain red rounded shapes, leading to confusion in retrieval order. This requires an explicit ranking strategy that not only identifies relevant local features but also maintains semantically consistent ordering across the entire candidate set.

To address the above challenges, we propose a **Dual-Branch Multi-Granularity network** with structured contrastive ranking (DBMG) for CMR. Our framework is designed to tackle semantic sparsity through multi-granularity alignment and resolve ranking inconsistency via a progressive optimization strategy. Specifically, we first introduce a multimodal large language model (MLLM) to generate auxiliary textual descriptions for images, enriching semantic representation and surfacing implicit visual cues. We then design a **dual-branch network** that captures both global and local interactions between images and texts, enabling fine-grained alignment across modalities. To further ensure robust ranking performance, we propose a **structured contrastive ranking loss** that operates in three complementary stages: (1) a category-aware contrastive loss that coarsely clusters samples from the same category; (2) an instance-level alignment loss that strictly enforces one-to-one correspondence between matched pairs; and (3) a category ranking margin loss that explicitly enlarges the relative distance between positive and negative samples. This hierarchical optimization strategy enables the model to not only identify sparse local cues but also maintain semantically consistent ranking across the candidate set. In summary, our contributions are shown as follows:

- We propose a novel dual-branch multi-granularity network that leverages MLLM-generated descriptions with a modular alignment architecture to address semantic sparsity, by capturing both explicit and implicit cross-modal cues at global and local levels.
- We design a structured contrastive ranking loss that progressively optimizes retrieval performance from category-level clustering to instance-level matching and final ranking calibration, resolving the inconsistency between fine-grained matching and ranking order.
- We conduct extensive experiments on four widely used CMR datasets. Results show that DBMG significantly outperforms existing approaches, achieving an average improvement of 15.91% in mAP and establishing a new state-of-the-art, while also demonstrating strong generalization under low-resource settings.

2 Related Work

Cross-Modal Retrieval. Previous studies on CMR can be broadly categorized into two groups: (1) global similarity methods based on a shared embedding space and (2) hybrid approaches that incorporate both global and local modeling. The former typically employs separate encoders to embed data from different modalities, followed by nearest neighbor search in a common vector space [35, 48]. However, these methods often perform poorly due to the semantic gap between modalities. Recent studies instead adopt attention mechanisms to model global-local associations between images and text. In particular, some methods use object detectors to locate image regions of interest and compute attention scores with textual tokens for fine-grained alignment [5, 51]. Nevertheless, the effectiveness of these methods is highly constrained by the accuracy and coverage of the underlying object detectors. LECCR [42] uses image descriptions generated by MLLM for cross-language CMR but is

limited by the quality of these descriptions. To this end, COTS [25] enhances fine-grained cross-modal interaction by predicting labels for masked image regions, while ALBEF [22] applies masked modeling on the text side to strengthen alignment. Although such methods improve local semantic perception, frequent masking operations may lead to suboptimal convergence and limit scalability on large-scale datasets.

Ranking Strategies. The effectiveness of ranking directly impacts the performance of CMR. Distance-based metrics alone often struggle with semantic ambiguity in real-world settings. Recent advances show that contrastive learning holds promise for improving multi-modal and multi-view alignment [12, 19]. Among them, TCL [46] introduces cross-modal, intra-modal, and local contrastive losses to jointly optimize semantic consistency and structural integrity. URL [7] decouples alignment and ranking, leveraging a differentiable ranking objective to enhance retrieval performance. PCME [9] models one-to-many relations with probabilistic embeddings and optimizes based on ranking quality, while ASCMR [50] adopts a density-aware strategy to improve robustness and reduce sample bias. Despite progress in representation learning and ranking optimization, these methods still exhibit limited generalization ability in scenarios with semantic sparsity or ambiguity.

3 METHODOLOGY

3.1 Overview

As illustrated in Figure 2, DBMG consists of two stages: (1) Semantic Alignment and (2) Ranking Optimization. *In the first stage*, we employ an MLLM to generate image descriptions and use a unified encoder to extract both global and local embeddings for text and images. The text interaction branch (TIB) and the cross-modal alignment branch (CAB) are then used to model intra-text and cross-modal semantic relationships at both global and local levels, achieving multi-grained semantic alignment. *In the second stage*, a novel structured contrastive ranking loss is designed to optimize the similarity matrix generated in the first stage, thereby enhancing the model’s ability to rank positive samples more accurately.

3.2 Theoretical Foundation

Problem Formulation. In the CMR task, the input consists of two modalities: an image set $\mathcal{V} = \{\mathbf{x}_i^v\}_{i=1}^N$ and a text set $\mathcal{T} = \{\mathbf{x}_i^t\}_{i=1}^N$, where $\mathbf{x}_i^v \in \mathbb{R}^{d_v}$ and $\mathbf{x}_i^t \in \mathbb{R}^{d_t}$ denote the i -th image and text input, respectively.

The goal of CMR is to retrieve the most semantically relevant sample from one modality given a query from the other. The task is formulated as a ranking problem, where the objective is to assign higher similarity scores to matched pairs than to mismatched ones. The model is trained by maximizing the log-likelihood of correctly retrieving the paired sample over the training set $\mathcal{D}$, with the objective defined as:

$$\theta^* = \arg \max_{\theta} \sum_{(\mathbf{x}_i^t, \mathbf{x}_i^v) \in \mathcal{D}} \log p_{\theta}(\mathbf{x}_i^v | \mathbf{x}_i^t, \mathcal{V}), \quad (1)$$

where θ^* denotes the learned model parameters. Afterwards, we compute $p_{\theta}(\mathbf{x}_i^v | \mathbf{x}_i^t, \mathcal{V})$ by calculating the similarity of the text to each image.

Theoretical Analysis. Semantic ambiguity leads to ranking uncertainty, where certain negative samples may obtain higher similarity scores than their corresponding positives. To quantify this phenomenon, we define the expected ranking risk as

$$\mathcal{R}(\theta) = \mathbb{E}_i \mathbb{E}_{(j,k) \in \mathcal{P}_i \times \mathcal{N}_i} [\mathbb{I}[S_{i,k}(\theta) \geq S_{i,j}(\theta)]] \quad (2)$$

where $\mathcal{P}_i$ and $\mathcal{N}_i$ denote the positive and negative sample sets of instance i , and $S_{i,j}$ represents the similarity between the i -th text and the j -th image. The indicator function $\mathbb{I}[\cdot]$ equals 1 if the condition inside is true and 0 otherwise. The term $\mathcal{R}(\theta)$ measures the probability that a negative sample is ranked higher than its positive counterpart. To mitigate this uncertainty, we design a structured contrastive loss $\mathcal{L}_s$ consisting of three components: (1) category-level contrastive loss $\mathcal{L}_1$, (2) instance-level alignment loss $\mathcal{L}_2$, and (3) category ranking margin loss $\mathcal{L}_3$. Specifically, $\mathcal{L}_1$ compresses intra-class variance and enlarges the inter-class mean gap $\Delta = \mu_P - \mu_N$ to improve category compactness. According to Chebyshev’s inequality, the misranking probability satisfies

$$\mathbb{P}[S_{i,k} \geq S_{i,j}] \leq \frac{\sigma_P^2}{\sigma_P^2 + (\Delta/2)^2} + \frac{\sigma_N^2}{\sigma_N^2 + (\Delta/2)^2} \quad (3)$$

where μ_P, μ_N are the mean similarities of positive and negative pairs, and σ_P^2, σ_N^2 denote their variances. Smaller variances or larger mean gaps directly reduce the ranking risk. $\mathcal{L}_2$ further enlarges Δ by enforcing the diagonal dominance of the similarity matrix S , i.e., $S_{i,i} \gg S_{i,j}, S_{k,i}$, thereby enhancing cross-modal discrimination. In addition, $\mathcal{L}_3$ constructs an upper bound for the misranking event as

$$\mathcal{R}(\theta) \leq \frac{1}{m} \mathbb{E}_{i,j,k} [m + S_{i,k} - S_{i,j}]_+ \quad (4)$$

where m is a fixed margin and $[\cdot]_+$ denotes the hinge operator. Minimizing $\mathcal{L}_3$ thus directly reduces the probability of ranking errors. Let $\theta_s = \arg \min_{\theta} \mathcal{L}_s(\theta)$ and $\theta_b = \arg \min_{\theta} \mathcal{L}_b(\theta)$ denote the optimal parameters obtained by the structured loss and a partial loss, respectively. Then we have

$$\mathcal{R}(\theta_s) \leq \Phi(\theta_s) \leq \Phi(\theta_b) \quad (5)$$

indicating that the proposed structured contrastive objective theoretically achieves a tighter upper bound on ranking risk, yielding more stable and discriminative retrieval performance.

3.3 Stage I: Dual-Branch Multi-Granularity Network for Semantic Alignment

Hierarchical Multi-view Feature Encoding. To enhance fine-grained semantic alignment between images and texts in CMR, we introduce image descriptions $\mathbf{x}^d$ generated by an MLLM as auxiliary information.

Given an input image and a concise natural language prompt, the MLLM automatically generates textual descriptions that capture the main objects, scene context, and their semantic relationships. Consequently, each training sample is represented as a triplet $(\mathbf{x}_i^v, \mathbf{x}_i^t, \mathbf{x}_i^d)$. We adopt a unified encoder, CLIP, to extract both global and local features for each sample, defined as follows:

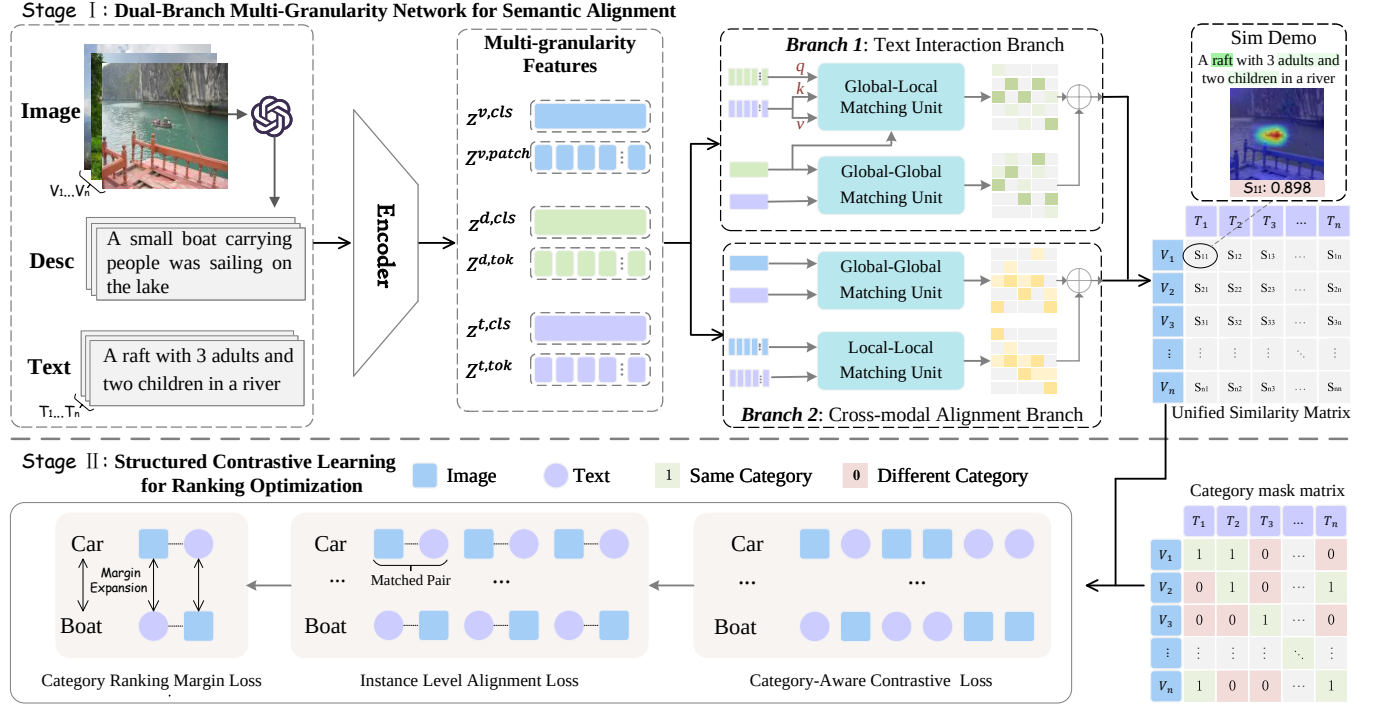


Figure 2: Overview of the DBMG framework. Stage I: MLLM is employed to generate auxiliary image descriptions. The image, its description, and the original text are encoded into global and local feature representations. The text interaction branch performs multi-granularity semantic matching between the original text and the generated image descriptions, producing two similarity matrices. Meanwhile, the cross-modal alignment branch models global and local correspondences between images and texts, generating another two similarity matrices. These outputs are then fused to construct the final text-image alignment matrix. Stage II: The fused alignment matrix is jointly optimized using structured contrastive ranking loss functions to enhance semantic accuracy and improve retrieval ranking performance.

$$\begin{aligned}
 z_i^{t,cls}, z_i^{t,tok} &= E_{\text{text}}(x_i^t), & z_i^{t,tok} &= \{z_{ij}^{t,tok}\}_{j=1}^{l_t} \\
 z_i^{d,cls}, z_i^{d,tok} &= E_{\text{text}}(x_i^d), & z_i^{d,tok} &= \{z_{ij}^{d,tok}\}_{j=1}^{l_d} \\
 z_i^{v,cls}, z_i^{v,patch} &= E_{\text{img}}(x_i^v), & z_i^{v,patch} &= \{z_{ij}^{v,patch}\}_{j=1}^{l_v}
 \end{aligned} \quad (6)$$

Here, z^{cls} denotes the global [CLS] feature representing the overall semantic content, while z^{tok} and z^{patch} refer to token-level textual features and patch-level visual features, respectively.

Dual-branch Multi-Granularity Semantic Matching. We design two parallel semantic interaction branches, as illustrated in Figure 2. The TIB focuses on modeling the semantic consistency between the original text and the generated image description, while the CAB directly aligns semantic information between text and images. Both branches model cross-modal semantic relationships at both global and local levels, enabling a more comprehensive understanding of multi-grained multimodal correspondences.

Text Interaction Branch consists of the global-to-global matching unit (TGG) and the global-to-local matching unit (TGL). The TGG measures the global semantic consistency between the original text and the image description by computing the cosine similarity between their [CLS] representations:

$$S^{TGG} = \cos(z^{t,cls}, z^{d,cls}) \quad (7)$$

To capture fine-grained semantic alignment, the Global-to-Local Matching Unit models the local-to-global interaction between the description tokens and the [CLS] embedding of the original text. Specifically, the token features of the description are projected as queries, while those of the original text are projected to keys and values:

$$Q = \text{Proj}(z^{d,tok}), \quad K, V = \text{Proj}(z^{t,tok}) \quad (8)$$

The attention output is computed as:

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (9)$$

We then apply mean pooling to the attended context and compute its cosine similarity with the projected description [CLS] embedding $\text{Proj}(z^{d,cls})$, yielding the local similarity score:

$$S^{TGL} = \left\langle \text{Proj}(z^{d,cls}), \text{mean}(\text{Attn}(Q, K, V)) \right\rangle \quad (10)$$

Finally, we combine the global-to-global and global-to-local similarity scores to obtain the overall semantic consistency between

the original text and the generated description. The final similarity score of the TIB is defined as their average:

$$S^{\text{TIB}} = \frac{1}{2} (S^{\text{TGG}} + S^{\text{TGL}}) \quad (11)$$

Cross-modal Alignment Branch consists of the global-to-global matching unit (CGG) and the local-to-local matching unit (CLL), which are used to compute the semantic correspondence between the original caption and the image. Although MLLMs help enhance the understanding of visual content, images still contain important visual cues that are difficult to fully capture by textual descriptions.

Specifically, the calculation processes of the two units are defined as:

$$\begin{aligned} S^{\text{CGG}} &= \cos \left(\text{Proj}(\mathbf{z}^{t,\text{cls}}), \mathbf{z}^{v,\text{cls}} \right) \\ S^{\text{CLL}} &= \max_{j,k} \cos \left(\text{Proj}(\mathbf{z}_j^{t,\text{tok}}), \text{Proj}(\mathbf{z}_k^{v,\text{patch}}) \right) \end{aligned} \quad (12)$$

where $\mathbf{z}^{t,\text{cls}}$ and $\mathbf{z}^{v,\text{cls}}$ denote the global [CLS] features of the text and image, respectively, while $\mathbf{z}_j^{t,\text{tok}}$ and $\mathbf{z}_k^{v,\text{patch}}$ represent the j -th token of the caption and the k -th patch feature of the image. $\text{Proj}(\cdot)$ indicates a linear projection into a shared embedding space. By computing the semantic relevance between each local region of the image and each word in the text, the model accurately captures semantically aligned information while effectively mitigating the influence of irrelevant or redundant content. The overall output of the CAB is defined as the average of the global and local similarity scores:

$$S^{\text{CAB}} = \frac{1}{2} (S^{\text{CGG}} + S^{\text{CLL}}) \quad (13)$$

The final cross-modal similarity score is obtained by aggregating similarity measures from two branches:

$$S(\mathbf{x}^t, \mathbf{x}^v) = S^{\text{TIB}} + S^{\text{CAB}} \quad (14)$$

3.4 Stage II: Structured Contrastive Learning for Ranking Optimization

Optimization Objective. To effectively supervise the proposed dual-branch, multi-granularity semantic alignment framework, we design a comprehensive optimization objective that addresses three key aspects: (1) semantic consistency at the category level, (2) precise instance-level text-image alignment, and (3) fine-grained ranking quality within semantic categories. Accordingly, we introduce three complementary loss functions to model these objectives individually and adaptively combine them using a learnable weighting strategy.

Category-Aware Contrastive Learning. To encourage samples from the same category to be close in the embedding space, we define a softmax-based contrastive loss over the similarity matrix. Let $\mathbf{S} \in \mathbb{R}^{B \times B}$ denote the pairwise similarity matrix computed between captions and images in a batch, and let $\tilde{\mathbf{S}} = \mathbf{S}/\tau$ be its temperature-scaled version, where τ is a temperature parameter. We introduce a category mask matrix $\mathbf{M} \in \{0, 1\}^{B \times B}$ as a category-matching indicator, where $M_{ij} = 1$ if sample i and j belong to the same class, and 0 otherwise. This mask is used to select positive pairs within the same category during contrastive learning. The text-to-image loss is formulated as:

$$\mathcal{L}_{\text{cat}}^{t \rightarrow v} = -\frac{1}{B} \sum_{i=1}^B \sum_{j=1}^B \frac{M_{ij}}{\sum_{k=1}^B M_{ik} + \epsilon} \log \left(\frac{e^{\tilde{S}_{ij}}}{\sum_{k=1}^B e^{\tilde{S}_{ik}}} \right) \quad (15)$$

where ϵ is a small constant to avoid division by zero. We also compute the symmetric image-to-text loss by applying the same formulation on the transposed similarity matrix. The final category-level contrastive loss is defined as:

$$\mathcal{L}_1 = \frac{1}{2} (\mathcal{L}_{\text{cat}}^{t \rightarrow v} + \mathcal{L}_{\text{cat}}^{v \rightarrow t}) \quad (16)$$

Instance Level Alignment Supervision. To enhance the model's ability to distinguish fine-grained semantic differences between individual samples, we introduce instance-level alignment supervision. Unlike the $\mathcal{L}_1$, instance-level alignment treats only strictly matched text-image pairs as positive samples, while all unmatched pairs are considered negatives. This loss maximizes the similarity of correctly paired samples and minimizes that of incorrect pairs, thereby improving the model's capacity to capture the unique semantic correspondence of each individual sample. It is defined as:

$$\mathcal{L}_{\text{ins}}^{t \rightarrow v} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp(S_{i,i})}{\sum_{j=1}^B \exp(S_{i,j})} \quad (17)$$

where B is the batch size, $\mathbf{S} \in \mathbb{R}^{B \times B}$ is the similarity matrix between texts and images, and $S_{i,j}$ denotes the similarity between the i -th text and the j -th image.

The final instance-level alignment loss is defined as the average of the text-to-image and image-to-text directions:

$$\mathcal{L}_2 = \frac{1}{2} (\mathcal{L}_{\text{ins}}^{t \rightarrow v} + \mathcal{L}_{\text{ins}}^{v \rightarrow t}) \quad (18)$$

Category Ranking Optimization. We propose a novel Category Ranking Margin Loss to enhance DBMG's ability to correctly rank positive samples. Specifically, for a given caption, the model is penalized if an image from a different category attains a higher similarity score than one from the same category. This margin-based loss is defined as:

$$\mathcal{L}_3 = \frac{1}{B} \sum_{i=1}^B \left[\frac{1}{|\mathcal{P}_i| |\mathcal{N}_i|} \sum_{j \in \mathcal{P}_i} \sum_{k \in \mathcal{N}_i} \max(0, m + S_{i,k} - S_{i,j}) \right] \quad (19)$$

where $\mathcal{P}_i$ and $\mathcal{N}_i$ are the sets of positive and negative image indices for caption i , and m is a fixed margin. Unlike $\mathcal{L}_1$, which encourages category-level clustering, $\mathcal{L}_3$ explicitly enforces the relative ranking between relevant and irrelevant samples, thereby directly contributing to improved top- k retrieval performance and higher mAP scores.

Structured Contrastive Ranking Loss. The three loss components are designed to complement each other: $\mathcal{L}_1$ promotes category-level semantic consistency, $\mathcal{L}_2$ enhances instance-level alignment between text and image pairs, and $\mathcal{L}_3$ optimizes the relative ranking between relevant and irrelevant samples, thereby improving CMR performance. To avoid manual tuning of weights, we adopt uncertainty-based loss balancing, and define the total training objective as:

$$\mathcal{L}_s = \sum_{i=1}^3 \left(\frac{1}{2\sigma_i^2} \mathcal{L}_i + \log \sigma_i \right) \quad (20)$$

Table 1: Performance comparison in terms of mAP scores on Wikipedia, Pascal Sentence, NUS-WIDE-10k, and XMediaNet.

Model	Source	Wikipedia			Pascal Sentence			NUS-WIDE-10k			XMediaNet		
		I→T	T→I	Avg.	I→T	T→I	Avg.	I→T	T→I	Avg.	I→T	T→I	Avg.
DSCMR	CVPR'2019	0.527	0.481	0.504	0.679	0.683	0.681	0.575	0.585	0.580	0.542	0.567	0.555
MAN	KBS'2019	0.528	0.480	0.504	0.680	0.700	0.690	0.565	0.575	0.570	0.537	0.550	0.544
ALGCN	TMM'2021	0.485	0.451	0.468	0.666	0.682	0.674	0.569	0.570	0.570	0.415	0.411	0.413
CAAL	TKDE'2022	0.507	0.463	0.485	0.568	0.592	0.580	0.542	0.553	0.548	0.587	0.575	0.581
ELRCMR	ACM MM'2022	0.543	0.501	0.522	0.705	0.700	0.702	0.549	0.567	0.558	0.300	0.283	0.292
MARS	TCSVT'2022	0.548	0.496	0.522	0.678	0.685	0.682	0.551	0.555	0.553	0.545	0.564	0.555
GNN4CMR	TPAMI'2023	0.521	0.480	0.501	0.705	0.701	0.703	0.595	0.591	0.593	0.523	0.538	0.531
RONO	CVPR'2023	0.521	0.473	0.497	0.711	0.701	0.706	0.559	0.590	0.575	0.456	0.464	0.460
SCL	TMM'2023	0.539	0.504	0.522	0.694	0.693	0.693	0.577	0.589	0.583	0.456	0.472	0.464
HOPE	TPAMI'2024	0.499	0.474	0.487	0.672	0.691	0.681	0.538	0.570	0.554	0.480	0.504	0.492
ICMR-DCT	ACM MM'2024	<u>0.607</u>	<u>0.596</u>	<u>0.602</u>	<u>0.767</u>	<u>0.753</u>	<u>0.760</u>	<u>0.624</u>	<u>0.626</u>	<u>0.625</u>	-	-	-
DRCL	TMM'2025	0.562	0.537	0.550	-	-	-	0.606	0.615	0.611	0.640	0.642	0.641
DBMG	Ours	0.667	0.633	0.650	0.856	0.854	0.855	0.808	0.802	0.805	0.729	0.737	0.733

where σ_i are learnable parameters associated with each loss term. This formulation enables the model to automatically adapt its learning dynamics across different granularities and modalities during training.

4 Experiments

4.1 Experimental Settings

Benchmarks. To comprehensively evaluate the performance of our proposed method DBMG, we compared it with a variety of state-of-the-art methods in the field. These methods include DSCMR [52], MAN [15], CAAL [14], ALGCN [30], MARS [41], ELRCMR [44], SCL [24], GNN4CMR [31], RONO [11], HOPE [49], ICMR-DCT [36], and DRCL [29].

Datasets and Evaluation Metrics. In our experiments, we adopt four widely used CMR benchmarks: Wikipedia [33], Pascal Sentence [32], NUS-WIDE-10k [8], and XMediaNet [28]. They contain 2,866, 1,000, 10,000, and 40,000 image-text pairs, respectively, covering diverse semantic categories from 10 to 200. We adopt mean Average Precision (mAP) as the evaluation metric to assess the performance of text-to-image and image-to-text retrieval tasks.

Implementation Details. Our model is implemented in PyTorch and uses the pretrained CLIP ViT-B/32 as the visual-textual encoder. Global and local features are projected into a shared 96-dimensional embedding space. The model is optimized with three loss functions, denoted as $\mathcal{L}_1$, $\mathcal{L}_2$, and $\mathcal{L}_3$. The temperature τ in $\mathcal{L}_1$ is set to 0.05, and the margin in $\mathcal{L}_3$ is set to 0.35. We use the AdamW optimizer with a learning rate of 1×10^{-5} and a weight decay of 1×10^{-2} , training the model for 50 epochs with a batch size of 32, except for XMediaNet, where the batch size is set to 128. Auxiliary image descriptions are generated by Qwen-VL-Plus prior to training and remain fixed throughout the learning process. All experiments are conducted on a Huawei Ascend 910B processor.

4.2 Comparison With State-of-The-Art Methods

Table 1 presents the comparative results, from which we answer the following Research Question (RQ).

RQ1 *How does the proposed DBMG improve the retrieval accuracy and generalization capability in CMR tasks compared with existing methods?*

We evaluate the proposed method on four widely used CMR datasets: Wikipedia, Pascal Sentence, NUS-WIDE-10k, and XMediaNet. A comprehensive comparison is conducted against 12 representative approaches. All baseline results are taken from publicly available, peer-reviewed publications to ensure fair comparison. As shown in Table 1, DBMG achieves the best performance across all datasets, demonstrating strong generalization ability across diverse multimodal scenarios. Compared with the state-of-the-art methods, it obtains average mAP improvements of 7.97%, 12.50%, 28.80%, and 14.35% respectively, significantly outperforming existing approaches.

Specifically, Pascal Sentence and NUS-WIDE-10K provide relatively concise textual inputs, including short descriptions and keyword-style category labels, respectively. Such text typically exhibits clear local semantic correspondence with salient regions in the image. By leveraging both global and local representations of image-text pairs, DBMG effectively captures fine-grained alignments through an attention mechanism between tokens and patches, resulting in more precise semantic matching.

In contrast, the textual content in Wikipedia and XMediaNet is often longer and more complex, containing extensive contextual or background information. The semantic relevance between text and image tends to be weaker, making local alignment more challenging. Nevertheless, DBMG maintains strong performance in these settings through its global semantic modeling and category-aware contrastive learning, which together mitigate the impact of redundant or loosely related content.

4.3 Ablation Study

To validate the effectiveness of each component in the proposed method, we conduct ablation experiments from two key perspectives: (1) the impact of different matching units on retrieval performance, and (2) the contribution of each loss function during training. Tables 2 and 3 present the experimental results that address the following research question:

RQ2 *How do the individual components and loss functions in DBMG enhance the performance of CMR?*

Impact of Each Component. To systematically analyze the contribution of each matching unit in DBMG, we conducted ablation

Table 2: Ablation study on matching units in Pascal Sentence. CGG and CLL denote two matching units in CAB, while TGG and TGL represent units from TIB. #2 indicates without MLLM-generated image descriptions.

No.	CAB		TIB		mAP
	CGG	CLL	TGG	TGL	
#1	✓				0.704
#2	✓	✓			0.808
#3	✓	✓	✓		0.822
#4			✓	✓	0.815
#5	✓	✓	✓	✓	0.855

experiments on the Pascal Sentence dataset, as shown in Table 2. The results show that as the four matching units are progressively incorporated, the retrieval performance continues to improve, indicating that both global and local matching mechanisms play crucial roles in achieving fine-grained semantic alignment between modalities. When only the cross-modal alignment branch (CAB, #2) and the text interaction branch (TIB, #4) are used, the model achieves mean Average Precision (mAP) scores of 0.808 and 0.815, respectively. Compared with the complete model (#5, mAP = 0.855), the performance decreases by 5.50% and 4.68%, respectively. These results demonstrate that the two branches are complementary within the overall framework.

Table 3: Ablation study on loss functions (mAP) on Wikipedia and NUS-WIDE-10k.

No.	$\mathcal{L}_1$	$\mathcal{L}_2$	$\mathcal{L}_3$	$\mathcal{L}_s$	Wikipedia	NUS-WIDE-10K
#1	✓				0.621	0.770
#2	✓	✓			0.632	0.781
#3	✓	✓	✓		0.644	0.791
#4	✓	✓	✓	✓	0.650	0.805

Impact of Loss Components. We evaluate the contribution of each loss component to retrieval performance on the Wikipedia and NUS-WIDE-10k datasets, as shown in Table 3. The results demonstrate that each loss term individually contributes to the overall performance of DBMG, while their joint optimization yields the best results, confirming their effectiveness and complementarity in improving CMR. Here, $\mathcal{L}_s$ denotes the loss fusion strategy, as detailed in Equation 20. Excluding $\mathcal{L}_s$ corresponds to simply combining the loss functions via unweighted addition.

4.4 Robustness Analysis

To further evaluate the performance of the proposed model under limited training data and analyze its dependency on the scale of the MLLM, we design two comparative experiments: (1) Low-Resource Setting, and (2) MLLM Scale Sensitivity Analysis. Figure 3 and Table 4 present the experimental results that address the following research question:

RQ3 *Is DBMG effective under limited training data, and to what extent does its performance depend on the scale of the MLLM?*

Low-Resource Experiment. In practical applications, collecting large-scale and high-quality labeled data often incurs substantial costs and significant time consumption. To evaluate the robustness of the proposed model under limited supervision, we train DBMG

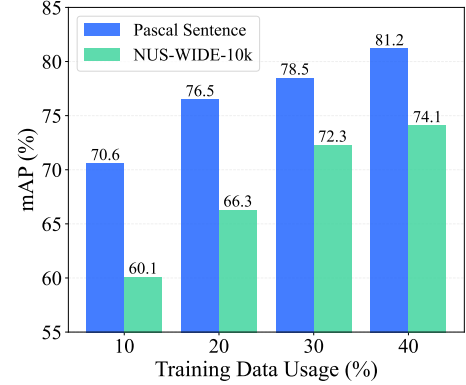


Figure 3: Performance of DBMG under varying training data proportions.

using only 10%–40% of the original training sets from the Pascal Sentence and NUS-WIDE-10k datasets. As illustrated in Figure 3, DBMG maintains strong performance even when the training data are greatly reduced. Notably, with only 20% of the training data, DBMG outperforms the strong baseline model ICMR-DCT on both datasets. These results demonstrate that DBMG exhibits higher data efficiency and stronger cross-modal generalization capability under data-constrained scenarios.

Table 4: Impact of different MLLMs on DBMG performance (mAP) on the Pascal Sentence.

Model	I→T	T→I	Avg.
DBMG (Qwen-2.5-VL-3B)	0.818	0.816	0.817
DBMG (Qwen-2.5-VL-7B)	0.854	0.848	0.851
DBMG (Qwen-2.5-VL-32B)	0.855	0.853	0.854
DBMG (Qwen-VL-Plus)	0.856	0.854	0.855

MLLM Scale Sensitivity Analysis. To evaluate the impact of MLLMs of different scales on DBMG, we compare its performance when integrated with various versions of Qwen-VL. As shown in Table 4, incorporating an MLLM significantly improves retrieval performance. Moreover, when adopting the lightweight Qwen-2.5-VL-7B, DBMG attains performance comparable to that of Qwen-VL-Plus. These results indicate that the proposed DBMG does not heavily rely on MLLMs.

4.5 Visualization and Analysis

In this section, we provide a visual analysis of DBMG’s performance from three perspectives: (1) coarse-grained feature distribution; (2) fine-grained feature alignment; and (3) retrieval ranking performance. Figures 4, 5, 6, and 7 present the experimental results that address the following research question:

RQ4 *Is the category-aware strategy in DBMG effective, and can it accurately capture sparse yet critical semantic cues from both modalities?*

Category-Aware Feature Distribution. We visualize the global features learned by DBMG using t-SNE [27], as shown in Figure 4. Samples from the same category form clear clusters in the shared embedding space. Some overlaps are observed in the image modality due to co-occurring concepts like “sky” and “cloud,” while the text

modality exhibits clearer separation. These results indicate that our learning strategy effectively promotes semantic alignment across modalities.

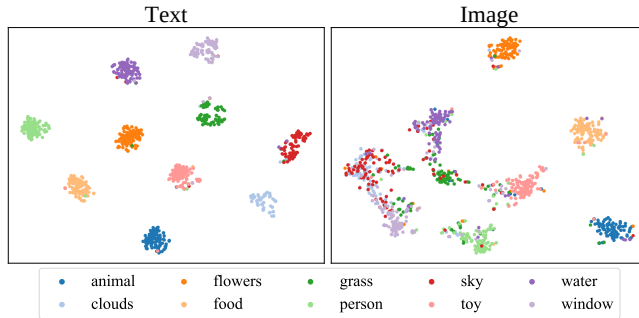


Figure 4: T-SNE visualization of samples from 10 categories in the NUS-WIDE-10k dataset. Samples from the same category share the same color.

Fine-Grained Cross-Modal Alignments. To enhance the interpretability of the proposed model, we employ Grad-CAM++ [4] to visualize the cross-modal retrieval behavior of DBMG, thereby revealing the model’s attention distribution over the image. As illustrated in Figure 5, the attention regions vary significantly with different query texts, where the color intensity indicates the degree of association between textual keywords and local image regions. In instance-level matching, the model’s attention is primarily concentrated on visual areas highly relevant to the textual content, demonstrating its strong capability to capture fine-grained semantic correspondences between visual and textual modalities.

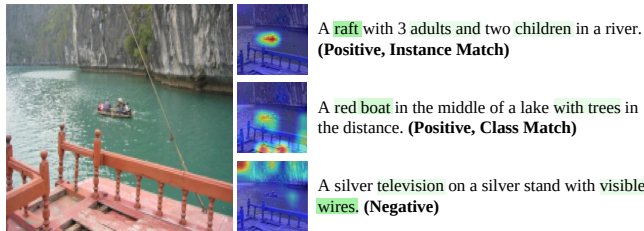


Figure 5: Grad-CAM++ visualization of cross-modal alignment between an input image and three retrieved texts: (1) an instance-level match, (2) a category-level match, and (3) a negative sample, showing that DBMG effectively captures semantically aligned regions.

Case Study. We further present a qualitative example to demonstrate the effectiveness of the proposed contrastive learning strategy. As shown in Figure 6, we illustrate four positive text samples from the same category as the image, along with one clear negative sample. Its lower similarity score indicates that DBMG effectively suppresses irrelevant candidates and mitigates ranking ambiguity in CMR. Furthermore, for the text-to-image retrieval task (as illustrated in Figure 7), we demonstrate DBMG’s ability to capture fine-grained semantic cues. In this example, all candidate images semantically match the query text “The plane soars in the sky,” yet the crucial clue “Austrian” appears only as textual information embedded within the image. Benefiting from the MLLM employed

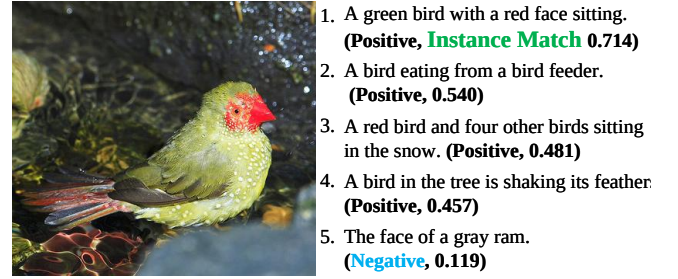


Figure 6: Image-to-text retrieval results on Pascal Sentence.

during the semantic alignment stage, DBMG can effectively recognize and leverage such embedded textual information, thereby achieving more precise cross-modal retrieval.

Text query: The Austrian plane soars in the sky.



Figure 7: Text-to-image retrieval results on Pascal Sentence.

5 Conclusion

In this work, we revisited the problem of cross-modal retrieval from the perspective of localized semantic evidence, revealing that both semantic sparsity and ranking inconsistency remain key obstacles to robust retrieval. Our study demonstrates that enriching visual semantics with MLLM-generated descriptions and explicitly aligning global and local cues can substantially narrow this gap. Beyond reporting strong empirical gains—achieving an average 15.91% improvement in mAP over 12 strong baseline methods—our findings highlight the importance of structured ranking objectives in maintaining retrieval consistency under semantic ambiguity. We further show that DBMG exhibits strong generalization under low-resource settings, indicating its robustness and practical applicability in scenarios with limited annotated data. Looking ahead, we plan to extend this framework to handle richer modalities, such as video [23, 45], paving the way toward more general-purpose multimodal reasoning systems.

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References

- [1] Mohammad Mahdi Abootorabi, Amirhosein Zobeiri, Mahdi Dehghani, Mohammadali Mohammadkhani, Bardia Mohammadi, Omid Ghahroodi, Mahdih Soleymani Baghshah, and Ehsaneddin Asgari. 2025. Ask in Any Modality: A Comprehensive Survey on Multimodal Retrieval-Augmented Generation. In *Findings of the Association for Computational Linguistics: ACL 2025*. Vienna, Austria, 16776–16809.
- [2] Azim Ahmadzadeh, Rohan Adhyapak, Armin Iraj, Kartik Chaurasiya, V. Aparna, and Petrus C. Martens. 2025. A Guide for Manual Annotation of Scientific Imagery: How to Prepare for Large Projects. *arXiv preprint arXiv:2508.14801* (2025). doi:10.48550/arXiv.2508.14801
- [3] Chenyang Bu, Guojie Chang, Zihao Chen, Cunyuan Dang, Zhize Wu, Yi He, and Xindong Wu. 2025. Query-Driven Multimodal GraphRAG: Dynamic Local Knowledge Graph Construction for Online Reasoning. In *Findings of the Association for Computational Linguistics: ACL 2025*. 21360–21380.
- [4] Aditya Chattopadhyay, Anirban Sarkar, Prantik Howlader, and Vineeth N. Balasubramanian. 2018. Grad-CAM++: Generalized Gradient-Based Visual Explanations for Deep Convolutional Networks. In *Proceedings of the IEEE Winter Conference on Applications of Computer Vision (WACV)*. 839–847.
- [5] Hui Chen, Guiguang Ding, Zijia Lin, Sicheng Zhao, Xiaopeng Gu, Wenyuan Xu, and Jungong Han. 2020. ACMNet: Adaptive confidence matching network for human behavior analysis via cross-modal retrieval. *ACM Transactions on Multimedia Computing, Communications, and Applications* 16, 1s (2020), 1–21.
- [6] Mengjun Cheng, Yipeng Sun, Longchao Wang, Xiongwei Zhu, Kun Yao, Jie Chen, Guoli Song, Junyu Han, Jingtuo Liu, Errui Ding, et al. 2022. VISTA: Vision and Scene Text Aggregation for Cross-Modal Retrieval. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 5184–5193.
- [7] Qingrong Cheng, Zhenshan Tan, Keyu Wen, Cheng Chen, and Xiaodong Gu. 2024. Semantic pre-alignment and ranking learning with unified framework for cross-modal retrieval. *IEEE Transactions on Circuits and Systems for Video Technology* 34, 7 (2024), 6503–6516.
- [8] Tat-Seng Chua, Jinhui Tang, Richang Hong, Haojie Li, Zhiping Luo, and Yantao Zheng. 2009. NUS-WIDE: A Real-World Web Image Database from the National University of Singapore. In *Proceedings of the ACM International Conference on Image and Video Retrieval (CIVR)*. 1–9.
- [9] Sanghyuk Chun, Seong Joon Oh, Rafael Sampaio de Rezende, Yannis Kalantidis, and Diane Larlus. 2021. Probabilistic Embeddings for Cross-Modal Retrieval. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 8415–8424.
- [10] Jianfeng Dong, Xirong Li, Chaoxi Xu, Xun Yang, Gang Yang, Xun Wang, and Meng Wang. 2022. Dual encoding for video retrieval by text. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 44, 8 (2022), 4065–4080.
- [11] Yanglin Feng, Hongyuan Zhu, Dezhong Peng, Xi Peng, and Peng Hu. 2023. RONO: Robust Discriminative Learning with Noisy Labels for 2D-3D Cross-Modal Retrieval. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 11610–11619.
- [12] Hang Gao, Cheng Liu, Zuosong Cai, Hongming Sun, Gaoyang Li, Ying Li, and Wei Du. 2025. A Novel Approach for Effective Partially View-Aligned Clustering with Triple-Consistency. *IEEE Transactions on Circuits and Systems for Video Technology* (2025).
- [13] Hao Guo, Xugong Qin, Jun Jie Ou Yang, Peng Zhang, Gangyan Zeng, Yubo Li, and Hailun Lin. 2025. Towards Natural Language-Based Document Image Retrieval: New Dataset and Benchmark. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 29722–29732.
- [14] Shiyuan He, Weiyang Wang, Zheng Wang, Xing Xu, Yang Yang, Xiaoming Wang, and Heng Tao Shen. 2022. Category alignment adversarial learning for cross-modal retrieval. *IEEE Transactions on Knowledge and Data Engineering* 35, 5 (2022), 4527–4538.
- [15] Peng Hu, Dezhong Peng, Xu Wang, and Yong Xiang. 2019. Multimodal adversarial network for cross-modal retrieval. *Knowledge-Based Systems* 180 (2019), 38–50.
- [16] Wenbo Hu, Jia-Chen Gu, Zi-Yi Dou, Mohsen Fayyaz, Pan Lu, Kai-Wei Chang, and Nanyun Peng. 2025. MRAG-Bench: Vision-Centric Evaluation for Retrieval-Augmented Multimodal Models. In *Proceedings of the Thirteenth International Conference on Learning Representations (ICLR)*. <https://openreview.net/forum?id=Uskli4gMc>
- [17] Manzong Huang, Chenyang Bu, Yi He, and Xindong Wu. 2025. How to Mitigate Information Loss in Knowledge Graphs for GraphRAG: Leveraging Triple Context Restoration and Query-Driven Feedback. In *Proceedings of the International Joint Conference on Artificial Intelligence (IJCAI)*.
- [18] Manzong Huang, Chenyang Bu, Yi He, Xingrui Zhuo, and Xindong Wu. 2026. Relink: Constructing Query-Driven Evidence Graph On-the-Fly for GraphRAG. *arXiv preprint arXiv:2601.07192* (2026).
- [19] Chao Jia, Yinfei Yang, Ye Xia, Yi-Ting Chen, Zarana Parekh, Hieu Pham, Quoc Le, Yun-Hsuan Sung, Zhen Li, and Tom Duerig. 2021. Scaling Up Visual and Vision-Language Representation Learning with Noisy Text Supervision. In *Proceedings of the 38th International Conference on Machine Learning (ICML)*. 4904–4916.
- [20] Dongwon Kim, Namyp Kim, and Suha Kwak. 2023. Improving Cross-Modal Retrieval with Set of Diverse Embeddings. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 23422–23431.
- [21] Chia-Wen Kuo and Zsolt Kira. 2022. Beyond a Pre-Trained Object Detector: Cross-Modal Textual and Visual Context for Image Captioning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 17969–17979.
- [22] Junnan Li, Ramprasaath Selvaraju, Akhilesh Gotmare, Shafiq Joty, Caiming Xiong, and Steven Chu Hong Hoi. 2021. Align before fuse: Vision and language representation learning with momentum distillation. *Advances in neural information processing systems* 34 (2021), 9694–9705.
- [23] Mengyuan Liu, Jinfu Liu, Yongkang Jiang, and Bin He. 2026. Heatmap Pooling Network for Action Recognition from RGB Videos. *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2026).
- [24] Yaxin Liu, Jianlong Wu, Leigang Qu, Tian Gan, Jianhua Yin, and Liqiang Nie. 2022. Self-supervised correlation learning for cross-modal retrieval. *IEEE Transactions on Multimedia* 25 (2022), 2851–2863.
- [25] Haoyu Lu, Nanyi Fei, Yuqi Huo, Yizhao Gao, Zhiwu Lu, and Ji-Rong Wen. 2022. COTS: Collaborative Two-Stream Vision-Language Pre-Training Model for Cross-Modal Retrieval. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 15692–15701.
- [26] Yongdong Luo, Xiaowu Zheng, Xiao Yang, Guilin Li, Haojia Lin, Jinfa Huang, Jiayi Ji, Fei Chao, Jiebo Luo, and Rongrong Ji. 2024. Video-rag: Visually-aligned retrieval-augmented long video comprehension. *arXiv preprint arXiv:2411.13093* (2024).
- [27] Laurens van der Maaten and Geoffrey Hinton. 2008. Visualizing data using t-SNE. *Journal of machine learning research* 9, Nov (2008), 2579–2605.
- [28] Yuxin Peng, Jinwei Qi, and Yuxin Yuan. 2018. Modality-specific cross-modal similarity measurement with recurrent attention network. *IEEE Transactions on Image Processing* 27, 11 (2018), 5585–5599.
- [29] Ruitao Pu, Yang Qin, Dezhong Peng, Xiaomin Song, and Huiming Zheng. 2025. Deep Reversible Consistency Learning for Cross-Modal Retrieval. *IEEE Transactions on Multimedia* 27 (2025), 4095–4106.
- [30] Shengsheng Qian, Dizhan Xue, Quan Fang, and Changsheng Xu. 2021. Adaptive label-aware graph convolutional networks for cross-modal retrieval. *IEEE Transactions on Multimedia* 24 (2021), 3520–3532.
- [31] Shengsheng Qian, Dizhan Xue, Quan Fang, and Changsheng Xu. 2022. Integrating multi-label contrastive learning with dual adversarial graph neural networks for cross-modal retrieval. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 45, 4 (2022), 4794–4811.
- [32] Cyrus Rashtchian, Peter Young, Micah Hodosh, and Julia Hockenmaier. 2010. Collecting Image Annotations Using Amazon’s Mechanical Turk. In *Proceedings of the NAACL HLT 2010 Workshop on Creating Speech and Language Data with Amazon’s Mechanical Turk*. 139–147.
- [33] Nikhil Rasiwasia, Jose Costa Pereira, Emanuele Coviello, Gabriel Doyle, Gert R. G. Lanckriet, Roger Levy, and Nuno Vasconcelos. 2010. A New Approach to Cross-Modal Multimedia Retrieval. In *Proceedings of the 18th ACM International Conference on Multimedia (MM)*. 251–260.
- [34] Monica Riedler and Stefan Langer. 2024. Beyond text: Optimizing rag with multimodal inputs for industrial applications. *arXiv preprint arXiv:2410.21943* (2024).
- [35] Jan Rupnik and John Shawe-Taylor. 2010. Multi-View Canonical Correlation Analysis. In *Proceedings of the Conference on Data Mining and Data Warehouses (SiKDD 2010)*. 1–4.
- [36] Dan Shi, Lei Zhu, Jingjing Li, Guohua Dong, and Huaxiang Zhang. 2024. Incomplete cross-modal retrieval with deep correlation transfer. *ACM Transactions on Multimedia Computing, Communications and Applications* 20, 5 (2024), 1–21.
- [37] Shezheng Song, Shan Zhao, Chengyu Wang, Tianwei Yan, Shasha Li, Xiaoguang Mao, and Meng Wang. 2024. A dual-way enhanced framework from text matching point of view for multimodal entity linking. In *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*. 19008–19016.
- [38] Yuanmin Tang, Jing Yu, Keke Gai, Jiamin Zhuang, Gang Xiong, Yue Hu, and Qi Wu. 2024. Context-i2w: Mapping Images to Context-Dependent Words for Accurate Zero-Shot Composited Image Retrieval. In *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, Vol. 38. 5180–5188.
- [39] Xueyao Wan and Hang Yu. 2025. MMGraphRAG: Bridging Vision and Language with Interpretable Multimodal Knowledge Graphs. *arXiv preprint arXiv:2507.20804* (2025).
- [40] Tianshi Wang, Fengling Li, Lei Zhu, Jingjing Li, Zheng Zhang, and Heng Tao Shen. 2024. Cross-modal retrieval: A systematic review of methods and future directions. *Proc. IEEE* 112, 11 (2024), 1716–1754.
- [41] Yunbo Wang and Yuxin Peng. 2022. MARS: Learning modality-agnostic representation for scalable cross-media retrieval. *IEEE Transactions on Circuits and Systems for Video Technology* 32, 7 (2022), 4765–4777.
- [42] Yabing Wang, Le Wang, Qiang Zhou, Zhibin Wang, Hao Li, Gang Hua, and Wei Tang. 2024. Multimodal LLM Enhanced Cross-Lingual Cross-Modal Retrieval. In *Proceedings of the 32nd ACM International Conference on Multimedia (MM)*. 8296–8305.
- [43] Peng Xia, Kangyu Zhu, Haoran Li, Tianze Wang, Weijia Shi, Sheng Wang, Linjun Zhang, James Zou, and Huaxiu Yao. 2025. MMed-RAG: Versatile Multimodal RAG

- System for Medical Vision-Language Models. In *Proceedings of the International Conference on Learning Representations (ICLR)*.
- [44] Tianyuan Xu, Xueliang Liu, Zhen Huang, Dan Guo, Richang Hong, and Meng Wang. 2022. Early-Learning Regularized Contrastive Learning for Cross-Modal Retrieval with Noisy Labels. In *Proceedings of the 30th ACM International Conference on Multimedia (MM)*. 629–637.
 - [45] Sheng Yan, Yong Wang, Xin Du, Hongchang Jin, and Mengyuan Liu. 2024. Improving Fine-grained Understanding for Retrieval in Human Motion and Text. *IEEE Signal Processing Letters* (2024).
 - [46] Jinyu Yang, Jiali Duan, Son Tran, Yi Xu, Sampath Chanda, Liqun Chen, Belinda Zeng, Trishul Chilimbi, and Junzhou Huang. 2022. Vision-Language Pre-Training with Triple Contrastive Learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 15671–15680.
 - [47] Xu Yuan, Liangbo Ning, Wenqi Fan, and Qing Li. 2025. mKG-RAG: Multimodal Knowledge Graph-Enhanced RAG for Visual Question Answering. *arXiv preprint arXiv:2508.05318* (2025).
 - [48] Xiaohua Zhai, Yuxin Peng, and Jianguo Xiao. 2012. Cross-Modality Correlation Propagation for Cross-Media Retrieval. In *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. 2337–2340.
 - [49] Fan Zhang, Hang Zhou, Xian-Sheng Hua, Chong Chen, and Xiao Luo. 2024. Hope: A hierarchical perspective for semi-supervised 2d-3d cross-modal retrieval. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 46, 12 (2024), 8976–8993.
 - [50] Huaiwen Zhang, Yang Yang, Fan Qi, Shengsheng Qian, and Changsheng Xu. 2025. Active Supervised Cross-Modal Retrieval. *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2025).
 - [51] Qi Zhang, Zhen Lei, Zhaoxiang Zhang, and Stan Z. Li. 2020. Context-Aware Attention Network for Image-Text Retrieval. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 3536–3545.
 - [52] Liangli Zhen, Peng Hu, Xu Wang, and Dezhong Peng. 2019. Deep Supervised Cross-Modal Retrieval. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 10394–10403.
 - [53] Xingrui Zhuo, Shirui Pan, Jiapu Wang, Gongqing Wu, Zan Zhang, Rui Li, Zizhong Wei, and Xindong Wu. 2025. Progressive Prefix-Memory Tuning for Complex Logical Query Answering on Knowledge Graphs. In *Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence (IJCAI)*. IJCAI, 3716–3724.
 - [54] Xingrui Zhuo, Jiapu Wang, Gongqing Wu, Shirui Pan, and Xindong Wu. 2025. Effective Instruction Parsing Plugin for Complex Logical Query Answering on Knowledge Graphs. In *Proceedings of the ACM Web Conference (WWW)*. ACM, 4780–4792.

Appendix

A Supplementary Experiments

A.1 Evaluation Metrics

To evaluate retrieval performance, we adopt mean Average Precision (mAP), which measures the average precision across different recall levels in the ranked retrieval results. For each query i , the Average Precision (AP) is computed as:

$$AP_i = \frac{1}{R_i} \sum_{j=1}^N \frac{r_j}{j} \cdot rel_j \quad (21)$$

where R_i is the total number of relevant items for the i -th query, r_j denotes the number of relevant items among the top- j retrieved results, and $rel_j = 1$ if the item at rank j is relevant, and 0 otherwise.

The Mean Average Precision (mAP) over all Q queries is then calculated as:

$$mAP = \frac{1}{Q} \sum_{i=1}^Q AP_i \quad (22)$$

In our experiments, we compute AP_i over the entire ranked list without truncation. We report mAP scores for both text-to-image (T→I) and image-to-text (I→T) retrieval tasks, as well as their average.

A.2 Prompt Design for MLLM

We designed prompt templates to guide the MLLM in generating image descriptions, as illustrated in Figure 8. The following prompts were used for different datasets:

Describe the content of the image briefly in one sentence.

Listing 1: Prompt used for Pascal Sentence, NUS-WIDE-10k, and XMediaNet.

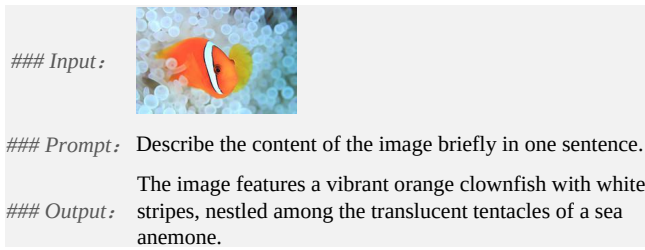


Figure 8: Example of the visual description generated using MLLM.

This prompt encourages the model to summarize the main content of the image concisely in a single sentence.

Briefly describe the content of the image based on its subject and background. The length must not exceed 77 tokens.

Listing 2: Prompt used for Wikipedia.

This prompt was used for Wikipedia. It adds an explicit length constraint and emphasizes both the subject and the background of the image.

A.3 Hyperparameter Analysis

In DBMG, the image encoder produces a 512-dimensional global feature and 768-dimensional local features. These are linearly projected into a shared embedding space for cross-modal alignment. In the main experiments, the embedding dimension is set to 96. To examine the effect of the embedding dimension on retrieval performance, we evaluate four projection dimensions: 48, 64, 96, and 128.

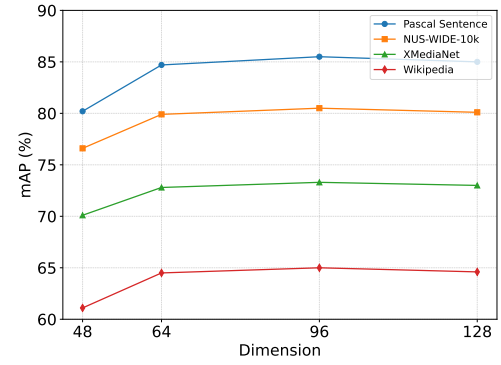


Figure 9: Impact of different embedding dimensions on DBMG performance.

As shown in Figure 9, the model achieves the highest mAP score when the embedding dimension is set to 96. A lower dimension restricts the model’s expressive capacity, while a higher dimension does not yield significant performance improvement and incurs greater computational cost. Therefore, we select 96 as the final embedding dimension.

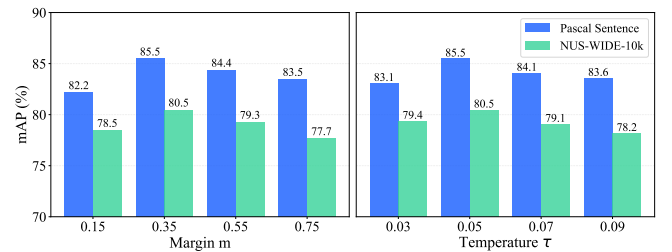


Figure 10: Impact of temperature and margin on cross-modal retrieval performance.

We further analyze two critical hyperparameters: the temperature coefficient τ in the loss function $\mathcal{L}_1$ and the margin m in $\mathcal{L}_3$. Experiments are conducted on the Pascal Sentence and NUS-WIDE-10K datasets. As shown in Figure 10, DBMG achieves the best retrieval performance when $\tau = 0.05$ and $m = 0.35$, which are thus used as default values in all experiments. All reported results in this subsection, as well as in the main experiments, are obtained by averaging over five runs.

A.4 Parameter and Baseline Analysis

Parameter Analysis. The proposed dual-branch multi-granularity model adopts CLIP as a unified image–text encoder to extract both global and local features from images and texts. On top of these representations, an interaction network is introduced to model cross-modal interactions at different granularity levels.

Table 5: Parameter comparison of CLIP-based retrieval models.

Model	Parameters
CLIP	151.40M
ICMR-DCT	156.14M
DBMG (Ours)	151.68M

As shown in Table 5, compared with the original CLIP model, DBMG introduces only an additional 0.28M parameters. Its overall model size is lower than that of the strong CLIP-based baseline ICMR-DCT, while improving retrieval performance without incurring noticeable parameter overhead.

CLIP-based Baseline Comparisons. To further isolate the contribution of the proposed architecture from the backbone and simple fusion strategies, we introduce two additional CLIP-based baselines under identical experimental settings. The first baseline (*CLIP-only*) computes retrieval scores using the global [CLS] token similarity with a standard contrastive loss. The second baseline (*CLIP+LLM (Naive)*) concatenates CLIP image features with MLLM-generated description embeddings (encoded by the same text encoder), followed by the same contrastive loss.

Table 6: Comparison with CLIP-based baselines under identical settings.

Method	Wikipedia	NUS-WIDE-10k	Pascal Sentence
CLIP-only	0.544	0.583	0.638
CLIP+LLM (Naive)	0.612	0.667	0.751
ICMR-DCT	0.602	0.625	0.760
DBMG (Ours)	0.650	0.805	0.855

As shown in Table 6, DBMG consistently outperforms the CLIP-only baseline, the naive CLIP+LLM fusion strategy, and the strong CLIP-based method ICMR-DCT across all datasets. These results demonstrate that the performance improvements of DBMG do not stem from the CLIP backbone or simple description fusion, but from the proposed dual-branch multi-granularity interactions and structured ranking objective.

B Empirical Validation of Ranking Risk

To complement the theoretical analysis in the main text, we empirically measure the key quantities involved in the ranking-risk bound, including the similarity gap between positive and negative pairs, the variance of similarity distributions, and the empirical ranking risk.

The empirical ranking risk is defined as the proportion of mis-ranked positive–negative pairs within a batch, corresponding to the expected risk in Eq. 2. The similarity gap (SimGap) reflects the mean difference $\Delta = \mu_P - \mu_N$ in Eq. 3, while PosVar and NegVar correspond to the variance terms σ_P^2 and σ_N^2 of the positive and negative similarity distributions, respectively.

Table 7 reports the results obtained by progressively introducing different loss components ($\mathcal{L}_1$, $\mathcal{L}_2$, $\mathcal{L}_3$, and adaptive weighting). We observe that each component consistently enlarges the similarity gap, reduces the variance of similarity distributions, and further decreases the empirical ranking risk. This trend is consistent with the theoretical conclusion that enlarging the mean gap and suppressing variance jointly reduce the upper bound of ranking risk, thereby validating the effectiveness of the proposed structured contrastive objective in practice.

Table 7: Empirical analysis of ranking-risk related quantities under different loss configurations.

Loss Configuration	SimGap $\uparrow$	PosVar $\downarrow$	NegVar $\downarrow$	Risk $\downarrow$
$\mathcal{L}_1$	0.689	2.054	2.362	0.115
$\mathcal{L}_1 + \mathcal{L}_2$	0.787	1.315	1.752	0.110
$\mathcal{L}_1 + \mathcal{L}_2 + \mathcal{L}_3$	1.251	1.141	1.133	0.103
Full DBMG (+Adaptive)	2.120	1.060	0.324	0.081

C Downstream Evaluation in Multimodal RAG

To evaluate the practical impact of DBMG beyond standard retrieval benchmarks, we further integrate DBMG as the retriever in a full multimodal retrieval-augmented generation (RAG) pipeline. Specifically, we adopt the recent MRAG-BENCH benchmark [16]; meanwhile, prior work has shown that retrieval accuracy is a strong predictor of final question-answering (QA) performance in RAG systems [16, 18].

In this setting, we replace the standard CLIP-based retriever in the Qwen-VL-Plus pipeline with our proposed DBMG, while keeping all other components unchanged. This controlled comparison allows us to directly assess the effect of improved retrieval quality on downstream QA tasks.

Table 8: Downstream QA performance when using different retrievers in a multimodal RAG pipeline.

Retriever	Partial Acc.	Temporal Acc.	Overall Acc.
CLIP (Qwen-VL-Plus)	0.442	0.604	0.523
DBMG (Ours)	0.516	0.705	0.611

As shown in Table 8, replacing the CLIP retriever with DBMG consistently improves QA accuracy across all evaluation categories. These results confirm that the retrieval gains achieved by DBMG effectively translate into improved end-task performance in practical multimodal RAG scenarios.